



MODULE 11

Embodied Cognition and Robotics

Can intelligence exist without a body?

ITAI 4374: Neuroscience as a Model for AI

Recap: Module 10

- Neural networks are powerful pattern matchers but brittle outside their training distribution.
- **Neuro symbolic AI** combines connectionist perception with symbolic reasoning for better generalization.
- **Global Workspace Theory** became an engineering blueprint: specialized modules sharing information through a central workspace.
- **Key takeaway:** How the brain thinks drives how we design AI architectures.



Module 11: Now we add the body.



PART 1

The Biology of Embodied Intelligence

What the body contributes to thought, movement, and learning

What Is Embodied Cognition?

The classical view: the brain is a central processor; the body is just input output hardware.

Embodied cognition: the body is an active participant in thinking not peripheral to it.

The shape of your hands shapes how you understand grasping. Having two eyes shapes your sense of depth. Moving through space shapes concepts like 'forward' and 'near.'

Intelligence is not a program running on a brain computer. It is a property of an organism acting in a world.



Embodied vs. Disembodied Intelligence



Input: Touch the object.

Output: Processing...

Output: Sensory feedback active...

Output: Feeling texture...

Output: Interaction complete.



Evidence: Intelligence Without a Central Brain



Octopus: ~500 million neurons two thirds in the arms, not the central brain. Each arm can sense, process, and react semi independently. The arms are cognitive agents.

Slime mold (*Physarum polycephalum*): single celled organism, no brain, no nervous system. Yet it solves mazes, finds shortest paths, and replicates city transit networks through body dynamics alone.

Implication for AI: intelligence can be distributed across a body it does not require a centralized processor.

The Cerebellum: The Brain's Movement Master

10% of total brain volume · **~80%** of all brain neurons
(~69 billion)

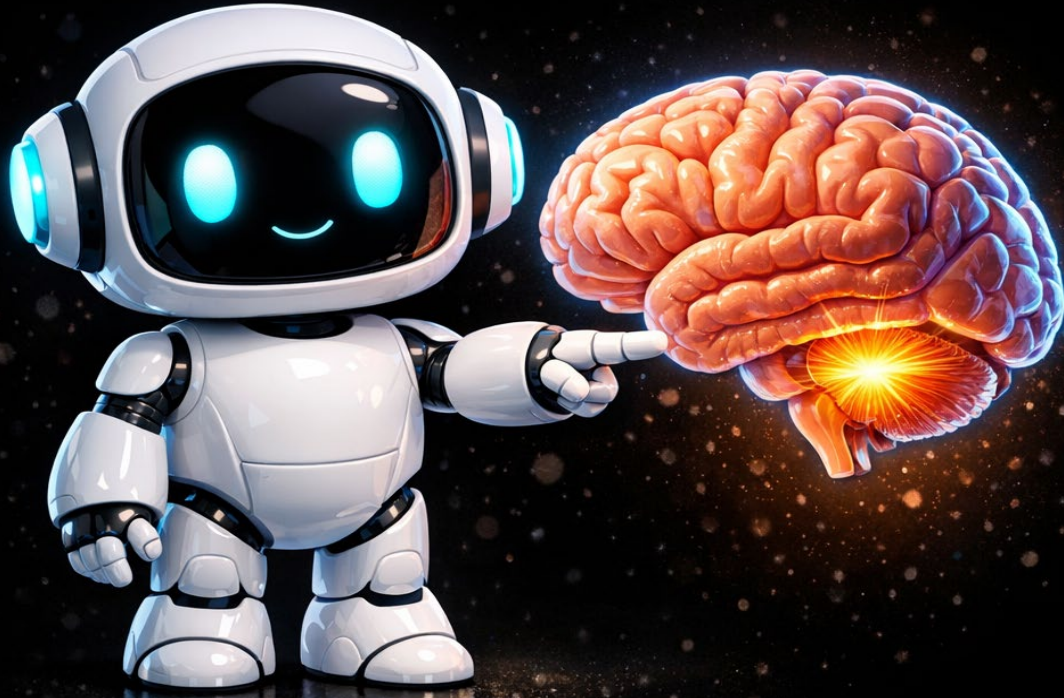
The cerebellum does NOT originate movement that is the motor cortex. It makes movement **precise, smooth, and well timed.**

Cerebellum vs. Basal Ganglia:

Basal Ganglia → WHAT to do (go/no go gating based on reward)

Cerebellum → HOW to do it (precision, timing, force)





Cerebellum Anatomy and Position

The Cerebellar Predict Act Correct Loop

① Motor cortex sends movement plan → to muscles AND to the cerebellum simultaneously.

② **Cerebellum predicts** what sensory feedback should arrive if the movement executes correctly.

③ Actual sensory feedback arrives from joints, muscles, tendons.

④ **Match:** no correction needed movement flows smoothly.
Mismatch: correction signal sent → model updated for the next attempt.

The cerebellum is fundamentally a learning machine, not just a movement machine.



Proprioception: The Hidden Sixth Sense

The continuous, largely unconscious sense of where your body parts are in space and how they are moving.

Three sensors:

Muscle Spindles → detect muscle length change and speed

Golgi Tendon Organs → detect muscle tension force

Joint Receptors → detect joint angle and movement direction

Interoception = the body's internal state (heart rate, hunger, temperature). ~80% of vagus nerve signals travel **upward** body informing brain.



Proprioceptive Sensors in the Human Body



The Perception Action Loop

Perception is not passive reception the brain continuously **generates predictions** about incoming sensory data, then compares predictions to what actually arrives.

The loop:

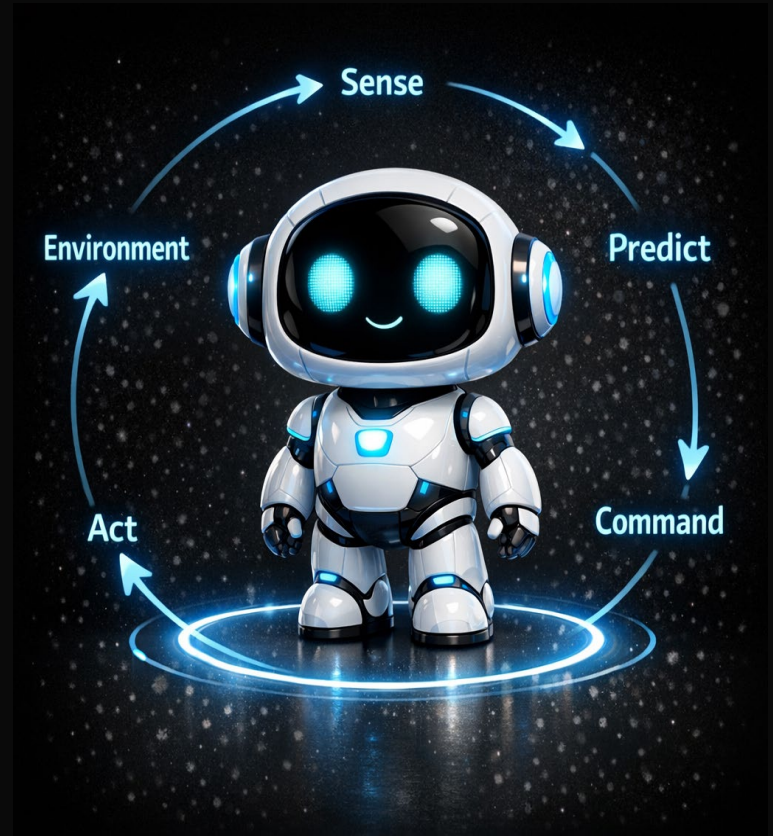
- Environment → Sense → Predict → Act → New sensory feedback → Update → Act again
- Every action is simultaneously a test of a prediction.

Efference Copy: when the brain sends a motor command, it simultaneously sends a copy to sensory regions so expected self-generated effects can be predicted and cancelled.

Try it: press gently on your closed eyelid. The world lurches no efference copy was sent.



The Perception Action Loop



Mirror Neurons: Observation = Action

Discovered at University of Parma (Gallese, Fadiga, Fogassi, Rizzolatti, 1996).

- **Mirror neurons** fire both when an animal **performs** a goal directed action and when it **observes** the same action.
- The same motor neuron same spike pattern regardless of who is doing the action.

Implications:

- → Babies imitate facial expressions within hours of birth
- → Watching expert performance activates your own motor patterns
- → Neural basis of empathy shares pathways with motor mirror systems



Mirror Neuron Activity: Observing vs. Performing



Imitation Learning in AI

- AI equivalent of the mirror neuron system: learning by observing demonstrations rather than trial and error.
- **Behavioral Cloning** directly copy observed actions. Simple. Brittle outside training scenarios.
- **Inverse Reinforcement Learning (IRL)** infer the goal behind demonstrated behavior, then pursue it independently. More generalizable.
- **Learning from Demonstration (LfD)** a human physically guides the robot through a task; the robot records and reproduces it. Programming through showing, not coding.



Industry: LfD has dropped robot programming time from days to hours for many manufacturing tasks.

Predictive Motor Control: Forward & Inverse Models

- When you pick up a coffee cup, your brain programs grip force from visual prediction before any weight information arrives from your hand.
- **Forward Model** → Given a motor command, predict the sensory outcome.
 - *"If I flex this much, my arm should rise by this angle in this time."*
- **Inverse Model** → Given a desired outcome, compute the motor command needed.
 - *"I want my hand there what muscle activations will get it?"*
- **Phantom limb pain:** forward model expects sensory confirmation that never arrives → stuck in an unresolvable prediction-error loop.





PART 2

Embodied AI Intelligence in the Physical World

*Affordances, sim to real, neuromorphic control,
and career frontiers*

What Changes When AI Gets a Body?

Frame Problem Solved Automatically: a physical agent does not need to reason about what changed its body filters relevant from irrelevant by its physical presence.

Affordances (Gibson, 1979): the action possibilities an environment offers to an agent with specific physical capabilities. A staircase affords climbing for a human not for a wheelchair user or a snake.

Morphological Computation: the body's physical structure performs computational work without neural processing.

Example: a kangaroo's tendons store and release kinetic energy at each hop, the nervous system computes nothing. The body solves the problem.



The Sim to Real Gap



The Sim to Real Gap

Simulation is fast, safe, and inexpensive.

Real environments are unpredictable, varied, and unforgiving.

The gap: policies trained in simulation degrade on real hardware because simulation cannot perfectly replicate friction, sensor noise, actuator wear, and lighting variation.

Domain Randomization (OpenAI DACTYL): deliberately vary simulation parameters during training lighting, friction, mass, noise so the real world is just another variation the agent has already encountered.

Training performance $\neq$ Real world performance one of the most important practical lessons in robotics.



Neuromorphic Control & Soft Robotics

Neuromorphic for Motor Control:

- **Ultra low latency** ~1ms spike response vs ~20ms GPU pipeline
- **Ultra low power** enables mobile, battery powered robot systems
- **On chip adaptation** continuous learning through sensorimotor experience

Soft Robotics (inspired by octopus, earthworm, jellyfish):

- flexible, compliant materials that deform on contact rather than breaking.
- **Applications:** agricultural harvesting · rehabilitation exosuits · surgical systems



Foundation Models for Robotics

What is a foundation model for robotics?

A single large AI model trained on thousands of different robot tasks so a robot can apply shared knowledge to new situations without being reprogrammed from scratch.

What is a Vision Language Action (VLA) model?

A model that combines three abilities in one: **see** (camera images), **understand language** (spoken or typed instructions), and **act** (output motor commands that move the robot).

Example Google DeepMind RT 2 (2023): *could pick up the 'healthiest food' from an unfamiliar table no specific training because it reasoned about health from language knowledge.*



The Embodied Turing Test

Turing's original test (1950): linguistic imitation game. It has a blind spot it ignores the physical world entirely.

Zador et al. (2023) propose a physical benchmark: can an AI system navigate, sense, and manipulate the physical world as competently as a biologically comparable organism?

The baseline is not a human it is a **mouse or rat**.

A mouse learns a novel maze in a few trials. Generalizes. Runs on a brain the size of a peanut at under 0.1 watts.

Current robots require thousands of trials, fail on modest variations, and consume hundreds of watts.



NVIDIA Isaac GROOT N Physical AI Pipeline



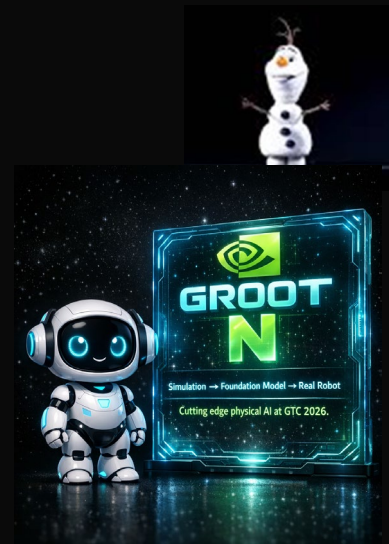
Isaac GROOT: NVIDIA's Physical AI Pipeline

NVIDIA's Physical AI pipeline Brain, sandbox, body. Built across two GTCs.

The brain: Isaac GROOT Open source robot foundation model. Launched at GTC 2025 with grasping, object handling, and voice commands pretrained. GTC 2026: N1.7 ships commercial, N2 previewed.

The sandbox: Newton + Isaac Sim Open source physics engine from NVIDIA, Google DeepMind, and Disney. Announced 2025, shipped as Newton 1.0 at GTC 2026, now plugged into Isaac Sim 6.0. Robots practice here before hitting the real world.

The body: real hardware Sim-trained models deploy to physical robots. GTC 2026: Disney's Olaf walked out with Jensen, headed for Disneyland Paris.



Module 11 Key Takeaway

Intelligence is not a program running on a brain computer.

It is a property of an organism acting in a world.

The body is not peripheral to cognition it shapes what the brain can perceive, what it decides, and what it learns. Embodied AI systems that ignore this gap will continue to fail where biological intelligence succeeds.



NEXT: Module 12

Consciousness and the Future of Brain Inspired AI

"Module 11 brought intelligence into the world. Module 12 asks the biggest question: what does it all mean?"

Coming up: Consciousness revisited · Hard Problem · IIT · GWT in embodied systems · Organoid intelligence · Where brain inspired AI is heading · What it means for your career

